AyurSamhita: A Healthcare Platform

A Minor Project Synopsis Submitted to

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal
Towards Partial Fulfillment for the Award of

Bachelor of Technology
(Computer Science and Engineering)

Under the Supervision of

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Jul – Dec 2025

1. Introduction of the Project

The Ayursamhita project is a digital healthcare initiative aimed at tackling the ongoing challenges that India's underserved communities face, including limited access to quality medical facilities, a shortage of doctors, and insufficient affordable healthcare infrastructure. It acknowledges the gap that exists between traditional systems like Ayurveda and modern medical practices, as well as the obstacles posed by India's extensive linguistic diversity. To address these challenges, the project proposes a platform that merges Ayurvedic knowledge with contemporary healthcare services, providing drug and formulation suggestions from classical texts, medical consultations from doctors, multilingual support, and collaboration among doctors. With this comprehensive approach, Ayursamhita seeks to deliver inclusive, dependable, and accessible healthcare, thus enhancing both preventive and curative care for various populations, especially in rural and underserved areas.

2. Objective

The Ayursamhita project aims to create a cutting-edge healthcare platform utilizing the MERN stack, merging the age-old wisdom of Ayurveda with the accuracy of contemporary medical practices. This initiative is designed to facilitate real-time consultations between patients and doctors, encourage collaboration among healthcare professionals, and offer multilingual support to promote inclusivity within various communities. By incorporating features like digital prescriptions, management of treatment histories, guidance for preventive care, and support for holistic wellness, the platform aspires to provide healthcare that is accessible, affordable, and easy to use. In the end, the objective is to establish a scalable, secure, and all-encompassing system that enhances integrated and holistic healthcare delivery.

3. Scope

The Ayursamhita project aims to create and launch a digital healthcare platform that merges Ayurvedic knowledge with contemporary medical practices into a cohesive system. This platform will offer consultations from doctor, support in multiple languages, and collaboration between doctors, particularly focusing on aiding rural and underserved populations. Notable features include recommendations for Ayurvedic medicines and formulations, digital prescriptions, and management of treatment histories to guarantee thorough care. However, it is important to note that the system is solely intended for healthcare guidance and support, and is not a substitute for licensed medical treatment. Therefore, it acts as a supportive, ethical, and user-friendly digital resource for delivering holistic healthcare.

4. Study of Existing System

To better understand the gaps AyurSamhita aims to fill, it's important to look at existing healthcare platforms. Each platform addresses various aspects of healthcare, but most fall short in integrating traditional Ayurvedic practices, multilingual support, and a focus on underserved communities.

1. **Practo:**
   - **Problems Addressed**: Online doctor consultations, appointment bookings, and health record management.
   - **Advantages**: Wide network of doctors, easy-to-use interface, and availability of both teleconsultations and in-clinic appointments.
   - **Disadvantages**: Lacks integration of traditional healthcare practices like Ayurveda and limited focus on rural healthcare.
   - **Gaps Identified**: No support for Ayurvedic consultations or multilingual assistance for rural populations.
   - **Reference Link**: https://www.practo.com

2. **1mg:**
   - **Problems Addressed**: Online consultations, medicine delivery, and lab test bookings.
   - **Advantages**: Medicine home delivery and lab test booking services with discounted rates.
   - **Disadvantages**: Limited focus on consultations and no integration of traditional medicine.
   - **Gaps Identified**: No Ayurvedic medical support and lacks comprehensive multilingual options for rural users.
   - **Reference Link**: https://www.1mg.com

3. **Lybrate:**
   - **Problems Addressed**: Online doctor-patient consultations across various specializations.
   - **Advantages**: Access to multiple specialists and real-time consultations.
   - **Disadvantages**: Primarily urban-focused and no integration of local languages or traditional medicine.
   - **Gaps Identified**: Lack of Ayurvedic practices and minimal focus on underserved communities.
   - **Reference Link**: https://www.lybrate.com

4. **Mfine:**
   - **Problems Addressed**: AI-powered health consultations and diagnostics.
   - **Advantages**: Quick consultations with specialists, AI-driven health advice, and online diagnostics.
   - **Disadvantages**: No traditional medicine integration and language support is limited to a few languages.
   - **Gaps Identified**: No Ayurvedic consultations or extensive multilingual support for rural areas.
   - **Reference Link**: https://www.mfine.co

5. **eSanjeevani:**

- **Problems Addressed**: Government-supported telemedicine platform offering free consultations.
- **Advantages**: Free consultations, wide reach across rural areas, and integration into government healthcare schemes.
- **Disadvantages**: Limited scope of specialized treatments and lacks integration with traditional Ayurvedic medicine.
- **Gaps Identified**: No Ayurvedic medicine integration and limited multilingual support for all regional languages.
- **Reference Link**: https://esanjeevani.mohfw.gov.in

5. Project Description

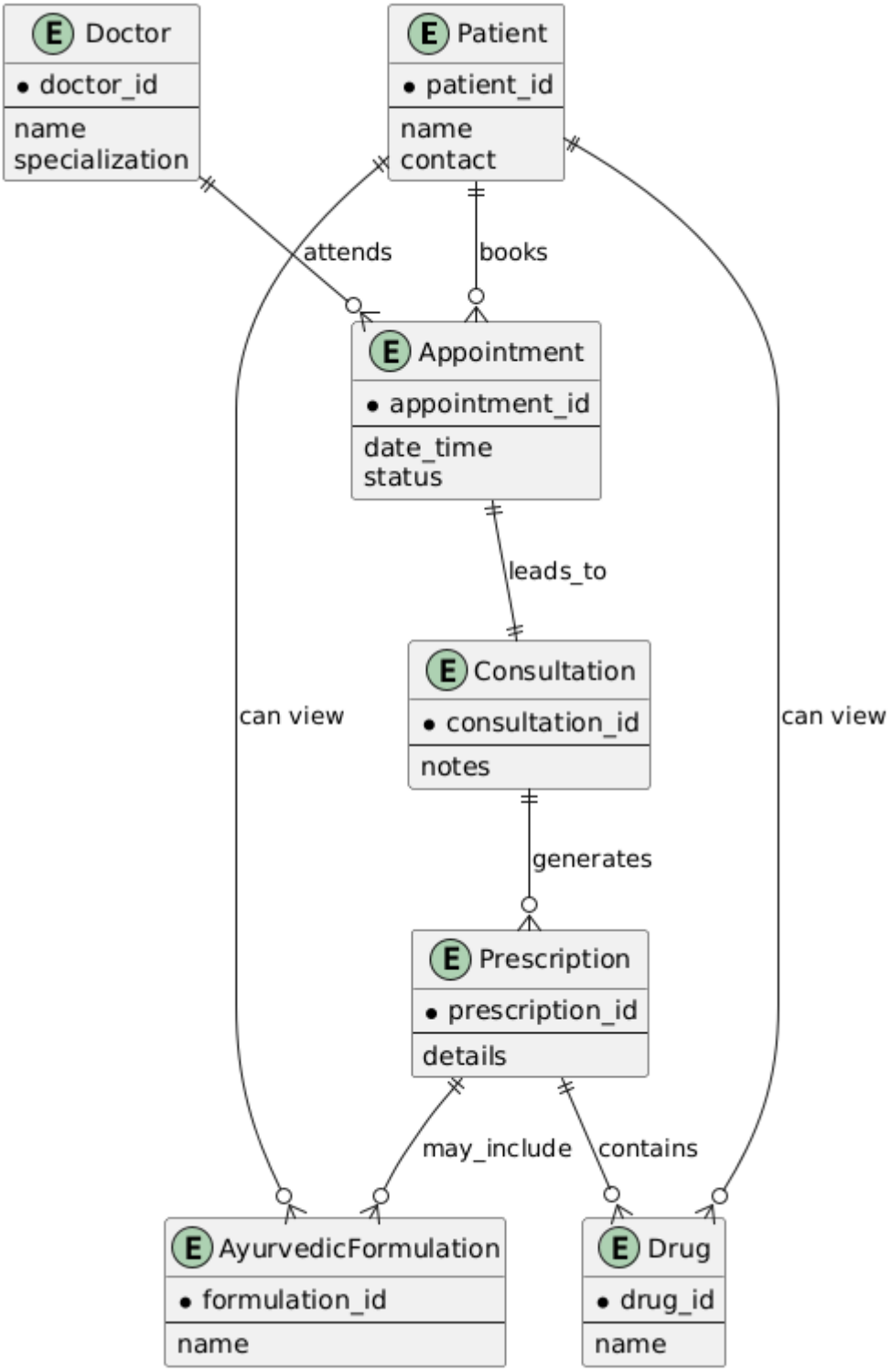
Ayursamhita is built on the MERN stack (MongoDB, Express.js, React.js, Node.js) to create a scalable, secure, and efficient healthcare platform. This project emphasizes the integration of Ayurvedic practices with contemporary medical consultations, offering patients a comprehensive approach to healthcare.

The platform facilitates collaboration among doctors, ensuring that specialists from both medical fields can work together for better decision-making and precise treatment recommendations. It also features video consultations from doctor, enabling patients to connect directly with healthcare providers, no matter where they are located.

To improve accessibility, the system includes multilingual support, making it user-friendly for people from various linguistic backgrounds, particularly in rural and underserved areas. The interface is designed to be simple and intuitive, allowing patients with limited technical skills to navigate it easily.

Additionally, the platform provides functionalities such as appointment scheduling, access to Ayurvedic formulations from traditional sources, and advice from modern healthcare professionals. By merging traditional knowledge with modern technology, Ayursamhita strives to deliver an inclusive, dependable, and patient-focused healthcare experience that connects underserved communities with quality healthcare.

We can see a very basic ER Diagram for the whole process.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Prescription
prescription_id
details
may_include
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formulation_id
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6. Planning of the Project work

The AyurSamhita platform's architecture is designed to ensure seamless integration between various components, providing a robust and scalable healthcare solution. The architecture facilitates real- time communication, secure data handling, and multilingual support, crucial for catering to India's underserved communities.

1. **Requirement Analysis (Weeks 1-2):** Conduct detailed research on user needs and existing platforms to define project scope, focusing on integrating Ayurvedic and modern medicine.

2. **System Design (Weeks 3-4):** Design the system architecture, including database schema, API endpoints, and UI wireframes, with a focus on multilingual and video consultation support.

3. **Backend and Database Development (Weeks 5-8):** Set up the backend using Node.js and Express.js, and implement a MongoDB database to manage user data, consultation records, and healthcare information.

4. **Frontend Development (Weeks 9-12):** Develop the user interface using React.js, ensuring a user-friendly experience for patients and doctors, and integrate more features such as dark mode, ayurvedic original text and educational section

5. **Integration and Testing (Weeks 13-16):** Integrate feature for video consultations from doctor to patient, and conduct comprehensive testing, including unit, integration, and user acceptance tests.

6. **Deployment (Weeks 17-19):** Making it available to user globally by deployment across environments, ensuring consistent performance and reliability.

AyurSamhita

GANTT CHART

7. Features

The development of AyurSamhita will follow an agile methodology, ensuring iterative progress and continuous feedback integration. The project will be divided into several key phases:

1. **Underserved Community Focus**: AyurSamhita is designed to provide equitable healthcare access to India's underserved communities, ensuring that everyone, regardless of location or language barriers, receives quality care. The platform prioritizes rural and marginalized populations by offering essential medical services in local languages.
2. **Doctor-to-Doctor Connect**: AyurSamhita facilitates seamless communication and collaboration among healthcare professionals, enabling a multidisciplinary approach to patient care. Doctors can consult with specialists across various fields, ensuring that patients receive comprehensive and well-rounded treatment.
3. **Ayurvedic Medicine Curation**: AyurSamhita offers consultations based on traditional Ayurvedic practices, curated by experienced practitioners. With a database covering 150+ diseases, the platform provides Ayurvedic solutions in local languages, promoting holistic and natural healing methods at the grassroots level.
4. **Video Consultations from doctors**:
   AyurSamhita leverages to offer video consultations between patients and healthcare providers. This feature enables personalized and interactive healthcare sessions, bridging the gap between rural populations and medical professionals.

8. System architecture

AyurSamhita is equipped with a range of innovative features designed to provide accessible and holistic healthcare to underserved communities in India. These features enable personalized care, traditional Ayurvedic practices, and real-time collaboration between healthcare professionals. Key features are:

1. **Users (Patients and Doctors):**
   Patients access the platform via a web or mobile interface built with React.js to book consultations, interact with the features, and view their medical records.
   Doctors use the platform to manage patient consultations, collaborate with other healthcare professionals, and access patient records.

2. **Frontend (React.js):**
   The frontend handles user interactions and sends requests to the backend via APIs. It includes features like video consultation booking, easy to go interactions, and multilingual support.

3. **Backend (Node.js + Express.js):**
   The backend processes user requests, manages data transactions with the MongoDB database, and provides API endpoints for frontend communication.
   It also handles connections for real-time video consultations and an extra section for those who need study of these ayurvedic texts (original and authentic).

4. **Database (MongoDB):**
   Stores patient information, consultation records, user profiles, and healthcare data securely. It supports scalable data handling, crucial for managing large volumes of user interactions.

5. **Video Consultations:**
   Facilitates real-time video consultations between doctors and patients, ensuring secure and smooth communication.

9. User Interface (UI)

The UI design will be simple yet effective, ensuring that users from rural backgrounds can navigate the platform with ease. Wireframes will be created to plan the layout of key pages such as the home screen, consultation booking, and doctor profiles. To enhance usability and comfort, the platform will also provide both dark mode and light mode options, allowing users to choose a visual theme that suits their preference and environment, thereby improving accessibility, reducing eye strain, and delivering a more personalized user experience.

10. Technology Stack

AyurSamhita leverages a powerful and diverse technology stack to deliver a robust healthcare platform tailored to India's underserved communities. By integrating modern web technologies, the platform ensures secure, scalable, and efficient healthcare solutions.

1. **Node.js**: Serves as the backend runtime environment for handling API requests, database interactions, and real-time communication.
2. **Vercel:** Provides a seamless deployment and hosting platform optimized for frontend applications, ensuring fast performance, global scalability, and automatic CI/CD integration. It simplifies the deployment process by offering serverless functions, edge networking, and real-time updates, allowing smooth delivery of the platform across different environments.
3. **Express.js**: Serves as the backend framework for building RESTful APIs and managing server-side routing.
4. **MongoDB**: Provides a NoSQL database solution for storing patient data, doctor profiles, consultation records, and user information.
5. **Tailwind CSS**: Used for building a responsive and intuitive front-end UI with custom styles that suit a healthcare platform.
6. **React.js**: Powers the front-end of the platform, providing a fast, scalable, and dynamic user interface for patients and doctors.

11. Testing Plan

To ensure AyurSamhita's functionality and reliability, the following straightforward testing methodologies will be implemented:

1. **Manual Testing:** Test critical features manually by navigating the platform to check for usability issues and basic functionality.

2. **Unit Testing:** Write and run simple tests for key functions (e.g., API endpoints, chatbot responses) using tools like Jest.

3. **Integration Testing:** Perform end-to-end tests by executing common tasks (e.g., booking consultations) to verify interaction between frontend and backend.

4. **Usability Testing:** Gather feedback from a small group of representative users, especially from rural areas, to assess ease of use and navigation.

5. **Performance Testing:** Simulate a few simultaneous users to check the platform's responsiveness under moderate traffic conditions.

12. Expected Outcome

AyurSamhita will deliver an accessible and inclusive healthcare platform that integrates traditional Ayurvedic medicine with modern practices. The platform will bridge the gap in healthcare access for underserved communities, providing real-time consultations and support for multiple languages. AyurSamhita will contribute to improving health outcomes for millions of people across India.

13. Resources and Limitations

Resources:

- **Hardware**: Servers for hosting consultations and handling user traffic.
- **Software**: MERN stack (MongoDB, Express js, React, Node js) with other libraries
- **Data**: Healthcare data and Ayurvedic medical resources for consultation services.
- **Personnel**: Doctors, Ayurvedic practitioners, and developers for maintaining the platform.

Limitations:

- **Internet Access**: Limited internet connectivity in rural areas could hinder real-time video consultations.
- **Adoption Barriers**: Educating underserved communities on using the platform may require additional outreach and training efforts.

14. Conclusion

AyurSamhita aims to revolutionize healthcare accessibility for India's underserved communities by integrating traditional Ayurvedic practices with modern medicine on a single platform. Leveraging the MERN stack and useful libraries and framework, the platform offers video consultations, doctor-to-doctor collaboration, and multilingual support to meet the diverse needs of rural populations. By bridging the gap between advanced healthcare and grassroots solutions, AyurSamhita ensures that quality healthcare is no longer restricted by geography or language barriers. Though challenges like internet connectivity and user adoption remain, the platform's holistic approach empowers communities with the tools necessary to access equitable healthcare. Ultimately, Arogya represents a forward-thinking solution that addresses the healthcare disparity in India, providing a scalable and inclusive model for the future.

15. References

[1]. "Introduction to Healthcare Informatics" – A comprehensive guide on the integration of IT in healthcare systems. [Author: Jane Smith]

[2] "Principles of Ayurvedic Medicine" – A foundational book on Ayurvedic practices and their integration with modern medicine. [Author: Ravi Kumar]

[3] V. Madaan and A. Goyal, "Predicting Ayurveda-based constituent balancing in human body using machine learning methods," IEEE Access, vol. 8, pp. 65060–65070, 2020.

[4] R. Vinuesa, H. Azizpour, I. Leite, M. Balaam, M. C. Chinnaiah, S. K. Dubey, N. Janardhan, V. R. Pathi, K. Nandan, and M. Anusha, "Analysis of Pitta imbalance in young Indian adult using machine learning algorithm," IEEE, published online June 24, 2022.

[5] IEEE 11073 SDC, "IEEE 11073 service-oriented device connectivity standard," IEEE Device Communication Standards, 2025. (Covers health device interoperability, relevant for telehealth modules)

[6] Simple and Compound Leaf Taxonomy Embedded Machine Learning Approach for Ayurveda Plants Recognition — 2024 4th International Conference on Advanced Research in Computing (ICARC), IEEE proceedings, DOI: 10.1109/ICARC61713.2024.10499741

[7] "Indian Science of Yajna and Mantra to Cure Different Diseases: An Analysis Amidst Pandemic with a Simulated Approach," included in Bioinformatics and Medical Applications: Big Data Using Deep Learning Algorithms (Wiley IEEE)

[8] "Designing Interfaces: Patterns for Effective Interaction Design" – A guide on UI/UX design principles for creating user-friendly interfaces. [Author: Jenifer Tidwell].